Marouane Daoudi

Montpellier, France · marouane.daoudi@etu.umontpellier.fr
marouanedaoudi.github.io · GitHub: marouanedaoudi · LinkedIn

Seeking a machine learning research internship in Taiwan.

Education

M2 Statistics and Data Science *2026–2027*
University of Montpellier, France · In progress
- Programme: statistical learning, Bayesian statistics, nonparametric estimation, generalized linear models, multivariate analysis and latent-variable models.

M1 Applied Mathematics *2025–2026*
University of Montpellier, France
- Probability and stochastic processes, statistical inference, optimization, time series, information and decision theory, multidimensional data analysis, linear models and software development.

Third year, Bachelor's degree in Mathematics *2024–2025*
University of Montpellier, France
- Measure theory, probability, stochastic modelling, convex optimization, numerical analysis, differential calculus and topology.

Scientific preparatory classes (CPGE) *2022–2024*
Lycée Descartes, Tours, France
- Intensive undergraduate preparation in mathematics, physics and computer science.

French general baccalaureate *2022*
Lycée Jean Moulin, Béziers, France · Mention Bien

Selected projects

Adaptive bandits and multiple testing Project · Code · Report, in French *2025–2026*
With Ines Bahraoui and Chams-Eddine Bouaziz; supervised by Adrien Nguyen-Huu and Luka Boisgibault.
- Study and implementation of sequential multiple-testing methods; comparison of adaptive and uniform allocation; simulations and analysis of statistical limitations.

L1BallPGD Project · Code *2025*
- NumPy implementations of PGD and FISTA for L1-constrained least squares, with exact L1-ball projection, synthetic experiments and numerical comparison with scikit-learn.

Slitherlink Shiny Project · Code · Documentation *2026*
- R package with grid validation, constraint propagation, backtracking, puzzle generation and an interactive Shiny interface.

Recip Score Project · Code *2024*
With Akram El Mahzoum, Chams-Eddine Bouaziz and Moussa Diagne.
- Python application combining recipe and nutritional data to compute scores and select recipes from available ingredients.

Technical skills

- **Python:** NumPy, pandas, scikit-learn; numerical computation and data analysis.
- **R:** dplyr, Shiny; statistical programming and package development.
- **Tools:** SQL, LaTeX, Git and GitHub.

Languages

English: C1 · Arabic: intermediate · Spanish: basic